
SSM Optimization

Weekly Meeting

24/07/26

Matmul

Input	$B \times D_1 \times D_2$
Weights	$D_2 \times D_3$
Output	$B \times D_1 \times D_3$

$$AI = \frac{FLOPs}{Bytes\ transferred} = \frac{2BD_1D_2D_3}{b (BD_1D_2 + D_2D_3 + BD_1D_3)}$$

with $b :=$ number of bytes used to represent a value
 $B :=$ batch size

Matmul

Let's consider $D_1 = D_2 = D_3 = 2048$, $b = 4$ (fp32):

Batch size	Theoretical			Profiled		
	FLOPs	Bytes transferred	AI	FLOPs	Bytes transferred	AI
1	17,179,869,184	50,331,648	341.33	17,209,229,312	33,631,488	511.70
4	68,719,476,736	150,994,944	455.11	68,736,253,952	126,270,592	544.36
8	137,438,953,472	285,212,672	481.88	137,472,507,904	259,857,920	529.03
16	274,877,906,944	553,648,128	496.49	274,945,015,808	526,469,760	522.24
32	549,755,813,888	1,090,519,040	504.12	549,890,031,616	1,066,726,272	515.49
64	1,099,511,627,776	2,164,260,864	508.03	1,099,780,063,232	2,146,367,104	512.39

Matmul

1. The profiled bytes transferred are less than the theoretical ones.
2. The offset between theoretical and profiled values for the transferred bytes is almost constant.

Batch size	Theoretical	Profiled	Gap
	Bytes transferred	Bytes transferred	
1	50,331,648	33,631,488	16,700,160
4	150,994,944	126,270,592	24,724,352
8	285,212,672	259,857,920	25,354,752
16	553,648,128	526,469,760	27,128,368
32	1,090,519,040	1,066,726,272	23,792,768
64	2,164,260,864	2,146,367,104	17,893,760

Probably imputable to the absence of the last write operation to DRAM, with the term being kept in L2

Matmul

Let's consider $D_1 = D_2 = D_3 = 2048$, $b = 4$ (fp32):

BS: 1

Measured FLOPs: 17,209,229,312

FADD instructions: 0

FMUL instructions: 12,582,912

FFMA instructions: 8,598,323,200

Each value of the output matrix ($B \times D_1 \times D_3$) is obtained with:

- $(D_2 - 1)$ FFMA
- 1 FMUL

However, this would mean that we should have:

- 8,585,740,288 FFMA
- 4,194,304 FMULs, which is exactly $1/3$ of measured FMULs

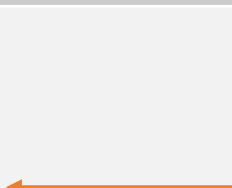
Probably
imputable to the
general form of
the matrix
multiplication
operation with
two constants

Matmul

Let's consider $D_1 = 2048$, $D_2 = 4096$, $D_3 = 16384$, $b = 4$ (fp32):

Batch size	Theoretical			Profiled		
	FLOPs	Bytes transferred	AI	FLOPs	Bytes transferred	AI
1	274,877,906,944	436,207,616	630.15	274,911,461,376	957,583,744	287.09
4	1,099,511,627,776	939,524,096	1170.29	1,099,645,845,504	2,754,870,656	399.17
8	2,199,023,255,552	1,610,612,736	1365.33	2,199,291,691,008	4,517,248,256	486.87
16	4,398,046,511,104	2,952,790,016	1489.46	4,398,583,382,016	7,775,983,360	565.66
32	8,796,093,022,208	5,637,144,576	1560.38	8,797,166,764,032	14,562,584,192	604.09
64	17,592,186,044,416	11,005,853,696	1598.44	17,594,333,528,064	27,452,456,064	640.90

Non-perfect weight reuse due to the dimensions of the weights matrix



Torch linear layer

Let's consider $D_1 = D_2 = D_3 = 2048$, $b = 4$ (fp32):

Batch size	Theoretical			Profiled		
	FLOPs	Bytes transferred	AI	FLOPs	Bytes transferred	AI
1	17,179,869,184	50,331,648	341.33	17,343,447,040	33,577,600	516.52
4	68,719,476,736	150,994,944	455.11	68,736,253,952	128,093,568	536.61
8	137,438,953,472	285,212,672	481.88	137,472,507,904	260,562,432	527.60
16	274,877,906,944	553,648,128	496.49	274,945,015,808	531,138,048	517.65
32	549,755,813,888	1,090,519,040	504.12	549,890,031,616	1,068,643,456	514.57
64	1,099,511,627,776	2,164,260,864	508.03	1,099,780,063,232	2,143,392,256	513.10

Torch linear layer

Let's consider $D_1 = 2048$, $D_2 = 4096$, $D_3 = 16384$, $b = 4$ (fp32):

Batch size	Theoretical			Profiled		
	FLOPs	Bytes transferred	AI	FLOPs	Bytes transferred	AI
1	274,877,906,944	436,207,616	630.15	274,911,461,376	456,599,040	602.09
4	1,099,511,627,776	939,524,096	1170.29	1,099,645,845,504	1,843,359,104	596.54
8	2,199,023,255,552	1,610,612,736	1365.33	2,199,291,691,008	3,801,138,304	578.59
16	4,398,046,511,104	2,952,790,016	1489.46	4,398,583,382,016	7,700,531,328	571.21
32	8,796,093,022,208	5,637,144,576	1560.38	8,797,166,764,032	15,137,942,272	581.13
64	17,592,186,044,416	11,005,853,696	1598.44	17,594,333,528,064	30,049,577,344	585.51

Falcon-Mamba-7B Mixer's in_proj

Let's consider $D_1 = 2048$, $D_2 = 4096$, $D_3 = 16384$, $b = 4$ (fp32):

Batch size	Theoretical			Profiled		
	FLOPs	Bytes transferred	AI	FLOPs	Bytes transferred	AI
1	274,877,906,944	436,207,616	630.15	2.749115e+11	4.559686e+08	602.92

Falcon-Mamba-7B Mixer (prefill)

Batch size = 1, seq_len = 2048, slow path

	fp32_cuda_core_flops	dram_bytes	arithmetic_intensity	gpu_time_ms
component				
in_proj	2.749115e+11	4.559686e+08	602.917476	14.950336
conv1d	1.342177e+08	2.315677e+08	0.579605	0.948896
x_proj	1.085997e+10	1.009466e+08	107.581383	0.632384
dt_proj	8.640266e+09	6.765901e+07	127.703117	0.556288
ssm	2.483172e+10	1.472713e+10	1.686121	62.403776
out_proj	1.374473e+11	6.964155e+08	197.363994	7.201152
mixer_overhead	3.550925e+08	1.140544e+08	3.113361	0.282880
falcon_mamba_mixer	4.571801e+11	1.639374e+10	27.887474	86.975712

Batch size = 1, seq_len = 2048, fast path

	fp32_cuda_core_flops	dram_bytes	arithmetic_intensity	gpu_time_ms
component				
in_proj	2.749115e+11	4.568794e+08	601.715651	14.954912
conv1d	4.865393e+08	9.337178e+07	5.210774	0.189472
x_proj	1.086849e+10	7.896282e+07	137.640611	0.710592
dt_proj	8.606712e+09	1.523685e+08	56.486158	0.905952
ssm	3.843555e+09	5.211209e+08	7.375554	2.198400
out_proj	1.374473e+11	4.229878e+08	324.944005	7.199168
mixer_overhead	2.770944e+06	7.404160e+06	0.374242	0.071680
falcon_mamba_fast_mixer	4.361669e+11	1.733095e+09	251.669295	26.230176

Batch size = 32, seq_len = 2048, fast path

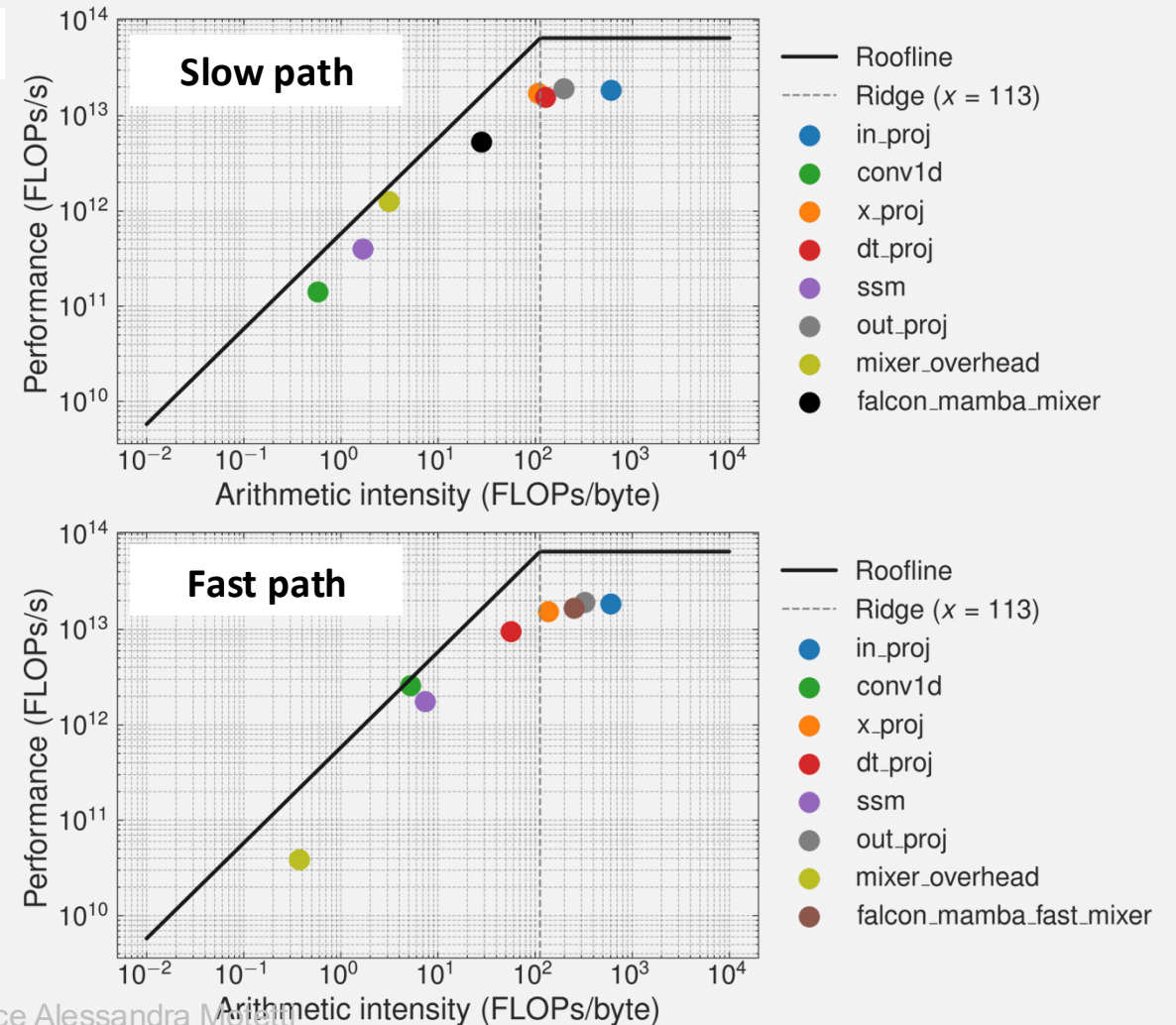
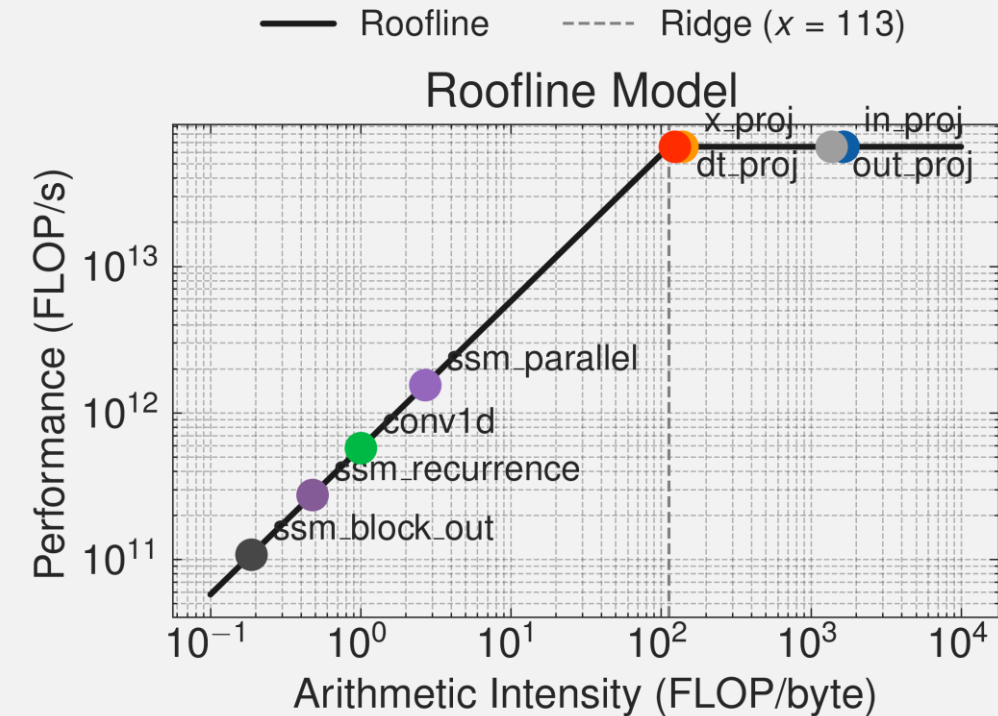
	fp32_cuda_core_flops	dram_bytes	arithmetic_intensity	gpu_time_ms
component				
in_proj	8.797167e+12	1.515641e+10	580.425480	466.065056
conv1d	1.556926e+10	4.262438e+09	3.652665	9.853696
x_proj	3.436184e+11	2.475967e+09	138.781449	19.745728
dt_proj	2.754148e+11	6.458534e+09	42.643545	28.377792
ssm	1.229491e+11	1.927139e+10	6.379877	68.819328
out_proj	4.398315e+12	2.267077e+10	194.008186	249.965376
mixer_overhead	8.867021e+07	2.848795e+08	0.311255	0.635616
falcon_mamba_fast_mixer	1.395312e+13	7.058039e+10	197.691207	843.462592

Falcon-Mamba-7B Mixer (decode)

Batch size = 1, slow path				
	fp32_cuda_core_flops	dram_bytes	arithmetic_intensity	gpu_time_ms
component				
in_proj	134512640.0	270510336.0	0.497255	0.511520
conv1d	245760.0	700800.0	0.350685	0.020384
x_proj	4728384.0	9486080.0	0.498455	0.023104
dt_proj	4726784.0	8428544.0	0.560807	0.020896
ssm	13590528.0	5730816.0	2.371482	0.065472
out_proj	67248128.0	136204672.0	0.493728	0.261280
mixer_overhead	204387.0	281600.0	0.725806	0.055648
falcon_mamba_mixer_decode	225256611.0	431342848.0	0.522222	0.958304

Falcon-Mamba-7B Mixer

Batch size = 1, seq_len = 2048, fp32



Profiling

Mamba-2 block

Batch size = 1, seq_len = 2048 (prefill)

	fp32_cuda_core_flops	dram_bytes	arithmetic_intensity	gpu_time_ms
component				
prenorm	3.591373e+07	20989824.0	1.711007	0.051456
in_proj	1.114225e+11	197894528.0	563.039682	6.253312
conv1d	1.211105e+08	53854336.0	2.299964	0.109984
ssm	2.112316e+08	156687104.0	16.552711	0.742944
gated_norm	1.730150e+08	100196736.0	1.726753	0.193152
out_proj	5.389156e+10	139359872.0	386.707901	3.057184
mamba2_customlib_block	1.658553e+11	668982400.0	251.487070	10.408032

Batch size = 1 (decode)

	fp32_cuda_core_flops	dram_bytes	arithmetic_intensity	gpu_time_ms
component				
prenorm	18944.0	25984.0	0.729064	0.004448
in_proj	54339488.0	110279936.0	0.492741	0.211488
conv_update	59136.0	201984.0	0.292776	0.004448
state_update	10650480.0	5382784.0	1.978619	0.036640
gated_norm	84480.0	76416.0	1.105528	0.007264
out_proj	26260480.0	52481280.0	0.500378	0.105664
mamba2_customlib_decode	91413008.0	168448384.0	0.542677	0.369952

Falcon-Mamba-7B Mixer: cast of state

State of the block casted to fp8, in a decoding setting:

Batch size = 1 (FP32, state FP8)

	fp32_cuda_core_flops	dram_bytes	arithmetic_intensity	gpu_time_ms
component				
in_proj	134512640.0	270433152.0	0.497397	0.511232
conv1d	245760.0	700928.0	0.350621	0.019936
x_proj	4728384.0	9486208.0	0.498448	0.023200
dt_proj	4726784.0	8428672.0	0.560798	0.021184
ssm	13703802.0	6427904.0	2.131924	0.075520
out_proj	67248128.0	136190336.0	0.493780	0.261408
mixer_overhead	204387.0	282240.0	0.724160	0.054656
falcon_mamba_mixer_decode_cast	225369885.0	431949440.0	0.521751	0.967136

Batch size = 1 (FP32)

	fp32_cuda_core_flops	dram_bytes	arithmetic_intensity	gpu_time_ms
component				
in_proj	134512640.0	270510336.0	0.497255	0.511520
conv1d	245760.0	700800.0	0.350685	0.020384
x_proj	4728384.0	9486080.0	0.498455	0.023104
dt_proj	4726784.0	8428544.0	0.560807	0.020896
ssm	13590528.0	5730816.0	2.371482	0.065472
out_proj	67248128.0	136204672.0	0.493728	0.261280
mixer_overhead	204387.0	281600.0	0.725806	0.055648
falcon_mamba_mixer_decode	225256611.0	431342848.0	0.522222	0.958304

Actually increases of 697,088, which could be due to the unpacking of the FP32 tensor in DRAM

Next steps

- Find a way to measure the impact of state compression on the FLOPs/DRAM transfers
 - If not possible at torch level, try to derive theoretical measures